I will give you screenshots of some handwritten math notes. You will explain me that topic in the way it is in the note and give intuition for better understanding. If it is a math or derivation, you will explain it step by step clearly. OK?

Okay, the user wants to provide screenshots for explanations. I can agree and guide them to upload the images. I'll set clear expectations by providing next steps: I'll annotate, explain key steps, discuss possible mistakes, and suggest alternative methods. I'll also ask if they want a template or typed notes/PDF.

To help with the image-based task, I'll ask the user to upload the necessary screenshots. I'll keep the response concise and clear, outlining how I'll structure the explanation: starting with an overview, followed by step-by-step reasoning. If needed, I'll clarify symbols, provide LaTeX/PDF, or create quizzes. Keeping a positive, warm tone!

Done

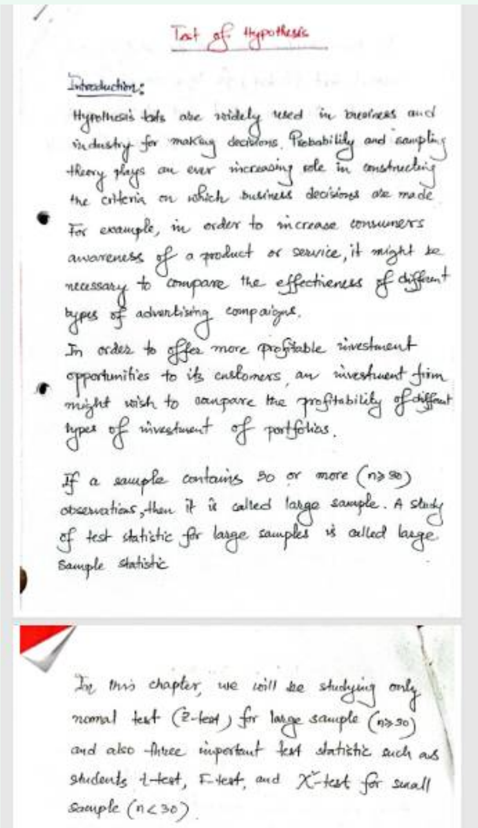
Absolutely! Send the screenshots whenever you're ready. Here's how I'll handle them:

- I'll **mirror the notation and order** exactly as in your notes.
- For any math/derivation, I'll go **line-by-line**, stating the rule used at each step (e.g., product rule, completing the square, variance formula expansion, etc.).
- I'll add **intuition** next to the algebra—why each step makes sense and how to remember it.
- I'll finish with a **quick recap**, common pitfalls, and a couple of **self-check questions**.

If parts are in Bangla, I can explain in **Bangla, English, or both**—your call.

Tip for clear photos: lay pages flat, good lighting, capture the whole page, and send them **in order** (1, 2, 3…).

Whenever you upload, I'll dive right in.



Test of Hypothesis

Introduction:

Hypothesis tests are widely used in business and industry for making decisions. Probability and sampling theory plays an ever increasing role in constructing the criteria on which business decisions are made.

- For example, in order to increase consumers awareness of a product or service, it might be necessary to compare the effectiness of different types of advertising campaigns.
- In order to offer more profitable investment opportunities to its customers, an investment firm might wish to compare the profitability of different types of investment of portfolios.

If a sample contains 30 or more ($n \geq 30$) observations, then it is called large sample. A study of test statistic for large samples is called large sample statistic

In this chapter, we will be studying only normal test (Z-test) for large sample ($n \geq 30$) and also three important test statistic such as students t-test, F-test, and χ^2-test for small sample ($n < 30$)

Great 👍 I can see your notes. Let's go through them carefully, step by step, and I'll add extra intuition so you fully understand the topic.

Topic: Test of Hypothesis

1. Introduction (from your notes)

- **What it says:**
  Hypothesis tests are widely used in business and industry for making decisions. Probability and sampling theory play an important role in forming criteria for decisions.
- **Intuition:**
  Imagine you're running a company. You can't ask every customer in the world what they think. Instead, you take a **sample** and use probability to decide whether your conclusion about the whole population is valid.
  → Hypothesis testing is a **decision-making tool under uncertainty**.

---

2. Examples given

1. **Advertising campaigns:**
   To check which advertisement works better (TV ad vs. online ad), we test hypotheses using customer awareness data.
2. **Investment opportunities:**
   An investment firm might compare the profitability of different types of portfolios. Again, we test hypotheses.

- **Intuition:**
  In real life, you always have multiple choices. Hypothesis testing gives you a way to **scientifically compare** options using data instead of just guessing.

---

3. Large vs. Small Sample

- **From the notes:**
  - If sample size **n ≥ 30** → **Large sample**.
  - If sample size **n < 30** → **Small sample**.
  - The test statistics are different for each.
- **Intuition:**
  - When the sample is large (≥30), by the **Central Limit Theorem (CLT)**, the sampling distribution of the mean becomes approximately **normal**, no matter what the population distribution is. So, we can use **Z-test**.
  - When the sample is small (<30), we cannot safely assume normality unless the population is normal. We then use **t-test, F-test, χ^2-test** depending on the situation.

---

4. From your notes: What this chapter covers

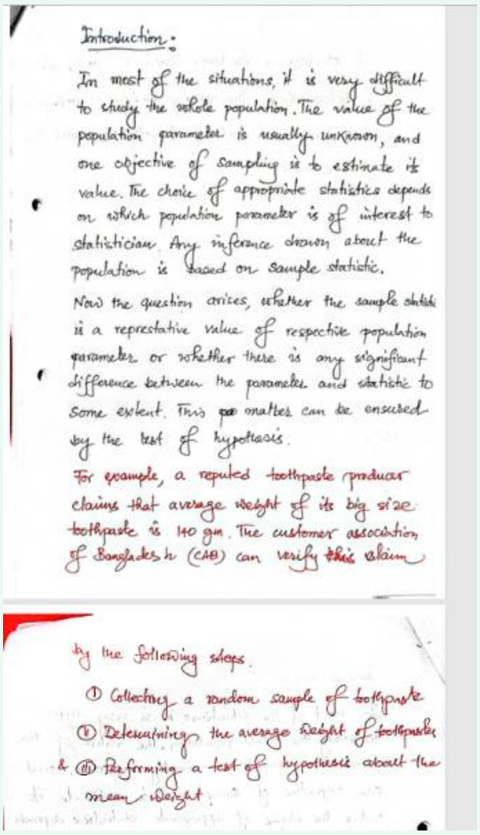
- For **large samples (n ≥ 30):** Z-test (normal test).
- For **small samples (n < 30):** t-test, F-test, χ^2-test.

✅ Quick Recap in Simple Words:

- Hypothesis testing = a structured way to make decisions using data.
- Large sample (≥30) → Z-test.
- Small sample (<30) → t, F, χ^2 tests.

- Applications: comparing ads, investments, medicine effectiveness, etc.

👉 Do you want me to next **explain the steps of hypothesis testing** (Null hypothesis, Alternative, Level of significance, Test statistic, Decision rule, Conclusion) in the same style as your notes?



Introduction:

In most of the situations, it is very difficult to study the whole population. The value of the population parameter is usually unknown, and one objective of sampling is to estimate its value. The choice of appropriate statistics depends on which population parameter is of interest to statistician. Any inference drawn about the population is based on sample statistic.

Now the question arises, whether the sample statistic is a representative value of respective population parameter or whether there is any significant difference between the parameter and statistic to some extent. This ~~po~~ matter can be ensured by the test of hypothesis.

For example, a reputed toothpaste producer claims that average weight of its big size toothpaste is 140 gm. The customer association of Bangladesh (CAB) can verify this claim by the following steps.

① Collecting a random sample of toothpaste

② Determining the average weight of toothpaste

& ③ Performing a test of hypothesis about the mean weight.

Perfect 👍 thanks for the next page. Let's carefully unpack it with **step-by-step explanation + intuition**.

Introduction (continued)

1. Whole Population vs. Sample

- **From notes:**
  It is usually difficult (almost impossible) to study the whole population. The population parameter (like mean μ or variance σ^2) is unknown.
  Sampling is done to estimate its value.

- **Intuition:**
  Imagine testing toothpaste weight. You cannot open and weigh every toothpaste tube from the factory (the *population*).
  Instead, you pick a **sample**, measure, and use it to **guess the truth about the whole population**.

---

2. Role of Sample Statistic

- **Notes:**
  The choice of statistic depends on which parameter is important (mean, proportion, variance, etc.).
  Any inference about the population is based on **sample statistics**.
- **Intuition:**
  - If you care about average weight → use **sample mean ($\bar{x}$)**.
  - If you care about percentage of defective items → use **sample proportion ($\hat{p}$)**.
    So: **Statistic = evidence**, **Parameter = truth**.

---

3. The Big Question

- **Notes:**
  Question: Is the sample statistic a good representative of the population parameter? Or is there a significant difference?
  → Answered using **Test of Hypothesis**.
- **Intuition:**
  Hypothesis testing is like a **court trial**:
  - Null hypothesis (H_0) = "company's claim is true."
  - Alternative (H_1) = "company's claim is false."
    The sample is like the **evidence**. Based on probability, we decide whether the evidence is strong enough to reject the claim.

---

4. Example (Toothpaste Claim)

- **Notes:**
  A toothpaste producer claims the average weight of big-size toothpaste is **140**

gm.
The **Consumer Association of Bangladesh (CAB)** wants to verify this claim.

- **Intuition:**
  H_0: $\mu = 140$ (company's claim is correct).
  H_1: $\mu \neq 140$ (claim is false).
  CAB will take a **sample of toothpaste tubes** to test this statistically.

---

5. Steps (as in your notes)

1. **Collect a random sample of toothpaste.**
   - (Avoid bias. Random sampling ensures fairness.)
2. **Calculate the average weight of toothpaste in the sample.**
   - This gives sample mean ($\bar{x}$).
3. **Perform a hypothesis test about the mean weight.**
   - Use Z-test (if $n \geq 30$) or t-test (if $n < 30$).
   - Compare test statistic with critical value.

✅ Summary in Simple Words:

- Population parameters are unknown → use sample statistics.
- We ask: "Is the sample evidence close enough to the claim?"
- Example: CAB checks if toothpaste really averages 140 gm.
- Procedure: Sample → Compute statistic → Test hypothesis.

👉 Do you want me to now show you **step-by-step procedure of hypothesis testing** (Null hypothesis, Alternative, Level of significance, Test statistic, Decision, Conclusion) with the toothpaste example worked out?

OpenAI

Definition of Hypothesis Testing:

The process that enables a decision-maker to draw an inference about population characteristics by analyzing the difference between the value obtained from sample and the hypothesized value of parameter is called hypothesis testing.

Concepts of Hypothesis Testing:

The process of hypothesis testing can be compared with criminal jury trial. In a jury trial it is assumed that the criminal is innocent, and the jury will decide that a person is guilty only if there is a very strong evidence against the presumption of innocence. This criminal jury trial process for choosing between guilt and innocence possesses

1. Rigorous procedures for presenting and evaluating evidence.

(II) A judge to enforce the rules;

(III) A decision process that assumes innocence unless there is evidence to prove guilt beyond a reasonable doubt.

It is to be noted that this process will fail to convict a number of people who are, in fact, guilty.

But if a person's innocence is rejected and the person is found guilty, we have a strong belief that the person is guilty. In testing the hypothesis, the sample collected from population acts as evidence of trial.

Perfect 👌 You've now shared the **definition and concept of hypothesis testing**. Let me break this down for you in **clear steps with intuition**.

Definition of Hypothesis Testing

- **From notes:**

> The process that enables a decision-maker to draw an inference about population characteristics by analyzing the difference between the sample value and the hypothesized parameter value is called hypothesis testing.

- **In simple words:**
  Hypothesis testing = Comparing **sample evidence** with a **claimed population value** to decide if the claim is believable.
- **Example:**
  A toothpaste company claims mean weight = 140 gm.
  We take a sample → If the average is *close enough* to 140, we accept the claim. Otherwise, we reject.

Concepts of Hypothesis Testing (Courtroom Analogy)

Your notes compare hypothesis testing to a **criminal jury trial**. This is a **classic and very helpful analogy**:

1. Presumption of innocence

- In a trial → Every person is assumed **innocent** until proven guilty.
- In hypothesis testing → Null hypothesis (**H_0**) is assumed **true** until there's strong evidence against it.

---

2. Rigorous procedures

- A trial has **strict rules** for presenting and evaluating evidence.
- Hypothesis testing has **mathematical rules** (significance level α, test statistic, decision rules).

---

3. Judge to enforce rules

- In a trial → Judge ensures fairness.
- In hypothesis testing → Statistical procedures ensure objectivity.

---

4. Strong evidence requirement

- In a trial → Jury convicts **only if evidence is beyond reasonable doubt**.
- In hypothesis testing → We reject H_0 **only if probability of sample evidence (p-value) is very small** under H_0.

5. Possibility of error

- Trials may wrongly acquit guilty people (Type II error) or wrongly convict innocent people (Type I error).
- Hypothesis testing has the **same risk**:
  - **Type I error (α):** Rejecting a true H_0 (convicting an innocent).
  - **Type II error (β):** Accepting a false H_0 (freeing the guilty).

---

6. The Role of Sample

- In trials → Evidence = witness, documents, facts.
- In hypothesis testing → **Sample data = evidence**.
- We use this evidence to decide whether to accept or reject H_0.

✅ Quick Summary (Court Analogy):

- **H_0 (Null Hypothesis)** = "Innocent until proven guilty."
- **Sample statistic** = "Evidence in court."
- **Significance level (α)** = "How strong the evidence must be."
- **Reject H_0** = "Convict the defendant."
- **Fail to reject H_0** = "Not enough evidence → defendant remains innocent."

👉 Do you want me to now explain **the formal steps of hypothesis testing (H_0, H_1, α, test statistic, decision rule, conclusion)** with a worked-out **toothpaste example** from your notes?

explain in more details and better intuition for the next ones

Got it ✅ — from here onward, I'll explain your notes **with much deeper details + crystal-clear intuition**.
I'll also keep connecting back to **real-world analogies** (like the courtroom one) so you don't just memorize, but actually *feel* why the steps of hypothesis testing make sense.

🔎 Hypothesis Testing — Deep Dive with Intuition

1. Definition (your notes restated)

> Hypothesis testing is the process of drawing inference about a population by comparing the sample result with the hypothesized population value.

🔑 Intuition:

- Think of it as **checking a claim**.
- The population claim is like a company's advertisement (e.g., "average toothpaste weight is 140 g").
- The **sample** is your reality check — like buying some tubes and weighing them.
- If sample results strongly disagree with the claim, you have reason to doubt it.

2. Courtroom Analogy Expanded (your notes)

This is one of the most powerful ways to "see" hypothesis testing.

Court Trial	Hypothesis Test
Accused is **innocent until proven guilty**	Null hypothesis (**H_0**) is assumed **true until evidence suggests otherwise**
Prosecutor presents **evidence**	We collect **sample data**
Jury compares evidence with the **burden of proof**	We compare test statistic with **critical value** (or p-value vs α)
Only if evidence is **very strong**, jury convicts	Only if sample result is **very unlikely under H_0**, we reject H_0
Errors happen: innocent may be jailed, guilty may go free	Errors happen: **Type I error** (reject true H_0), **Type II error** (accept false H_0)

3. Key Elements of the Analogy (from your notes + intuition)

1. **Rigorous procedures for presenting and evaluating evidence**
   - Courts have strict rules.

- Hypothesis testing uses precise formulas & probability rules → Z, t, F, χ^2.

2. **Judge to enforce rules**
   - In trials, the judge keeps order.
   - In statistics, the **significance level α** (e.g., 5%) keeps order.
   - You must stick to α before starting (no cheating by changing rules after seeing data).
3. **Decision process assumes innocence until proven guilty**
   - Courts → "innocent by default."
   - Statistics → "H_0 true by default."
   - We only overturn this assumption if evidence (sample result) is **too unlikely under H_0.**

---

4. Important Insight: Errors in Both Worlds

⚖️ In a trial:

- Sometimes guilty walk free (not enough evidence) → society suffers.
- Sometimes innocent are convicted (false conviction) → individual suffers.

📊 In hypothesis testing:

- **Type I error (α):** Rejecting H_0 even though it's true.
  - Like convicting an innocent.
  - Controlled by choosing α (usually 5% or 1%).
- **Type II error (β):** Not rejecting H_0 even though it's false.
  - Like freeing a guilty person.
  - Controlled by sample size and test design.

Intuition tip:

- Type I error = *"false alarm."*
- Type II error = *"missed detection."*

---

5. Connecting Back to the Toothpaste Example

- **H_0:** Mean toothpaste weight = 140 g (company claim is true).
- **H_1:** Mean ≠ 140 g (claim is false).

The **Consumer Association** acts like a **jury**:

- They collect **sample tubes** = evidence.
- They calculate **sample mean ($\bar{x}$)**.
- They ask: *"If true mean = 140, is our sample mean very unlikely?"*
- If yes → reject H_0 (company guilty of false claim).
- If no → fail to reject H_0 (not enough evidence, claim stands).

✅ **Super Intuitive Recap:**

- Hypothesis testing = a **trial of claims**.
- Null hypothesis = *innocent until proven guilty*.
- Sample = *evidence*.
- Significance level α = *burden of proof*.
- Reject H_0 only if evidence is **beyond reasonable doubt**.

👉 Next in your notes will likely be the **formal steps of hypothesis testing** (Null, Alternative, α, Test Statistic, Decision Rule, Conclusion).
Would you like me to go ahead and explain those **step by step with the toothpaste example worked out** (with formulas + intuition)?

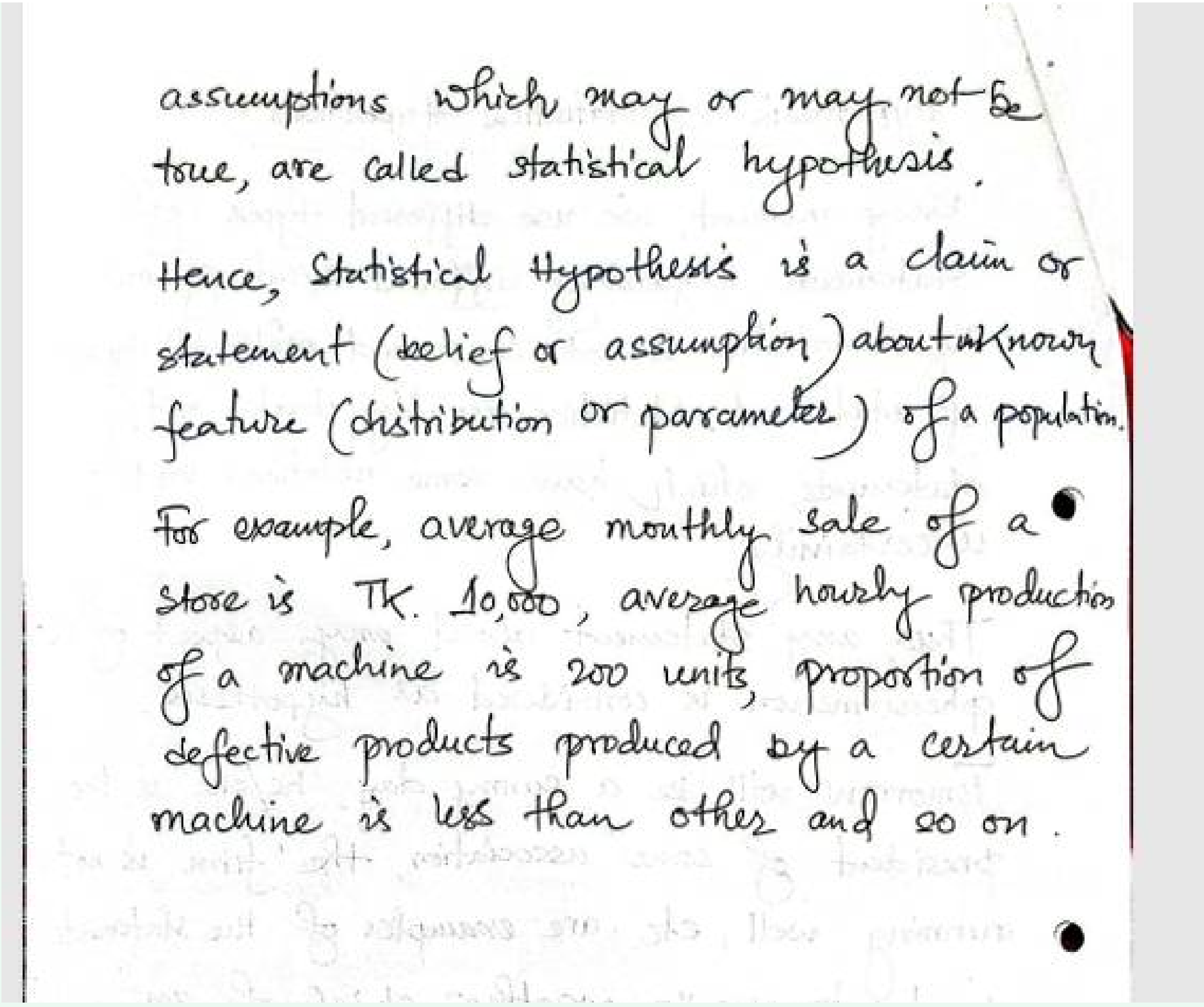
Hypothesis & Statistical Hypothesis :

Every moment, we use different types of statements regarding different events of our life. Statistics is not concerned with all types of statements. Statistics usually deals with statements, which have some relation with uncertainity.

Thus, any statement about any aspect of a phenomenon is considered as hypothesis.

Tomorrow will be a sunny day, he/she is the president of some association, the firm is not running well, etc. are examples of the statements about tomorrow's weather, chief of an association, state of the firm, respectively. Hence, these are simple hypothesis.

However, in attempting to take decision regarding some characteristics of population on the basis of sample, it is necessary to make some assumption regarding parameters of the population, such



assumptions which may or may not be true, are called statistical hypothesis.

Hence, Statistical Hypothesis is a claim or statement (belief or assumption) about unknown feature (distribution or parameter) of a population.

For example, average monthly sale of a store is TK. 10,000, average hourly production of a machine is 200 units, proportion of defective products produced by a certain machine is less than other and so on.

Perfect 🙌 You've now shared the section on **Hypothesis & Statistical Hypothesis**. Let's go through it step by step with **clear explanation + intuition**.

🔎 Hypothesis & Statistical Hypothesis

---

1. Hypothesis (General Idea)

- **From notes:**
  We constantly make statements about events in life. Not all of them are statistical, but many involve uncertainty.
  Example:
  - *"Tomorrow will be sunny."*
  - *"He is the president of the association."*
  - *"This firm is not running well."*

These are **simple hypotheses** in daily life.

- **Intuition:**
  A *hypothesis* is basically an **assumption or guess** about reality.
  - Some are about the future (weather).
  - Some are about facts (leadership).
  - Some are about conditions (company status).
    They all involve a claim, which could be true or false.

---

2. From Hypothesis → Statistical Hypothesis

- **From notes:**
  When we take decisions about **population characteristics** based on a **sample**, we need assumptions about the parameters (like mean, variance, proportion). These assumptions may or may not be true.
  → Such assumptions are called **Statistical Hypotheses**.
- **Intuition:**
  A **statistical hypothesis** is just a **claim about a population**.
  But since the whole population is usually unknown, we check this claim using sample data.

Think of it like:

- Normal hypothesis = "It might rain tomorrow."
- Statistical hypothesis = "The average rainfall in Dhaka in July is 350 mm."

---

3. Definition (your notes)

> A **statistical hypothesis** is a claim or statement (belief/assumption) about some unknown feature (distribution or parameter) of a population.

---

4. Examples from Notes (explained with intuition)

1. **"Average monthly sale of a store is Tk. 10,000."**
   → Claim about population mean ($\mu = 10{,}000$).

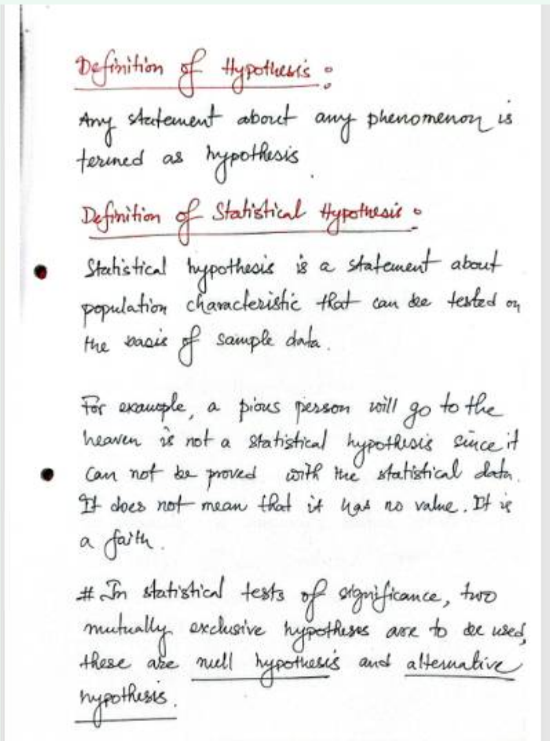
2. **“Average hourly production of a machine is 200 units.”**
   → Claim about production mean ($\mu = 200$).
3. **“Proportion of defective products produced by a certain machine is less than others.”**
   → Claim about population proportion ($p <$ something).

- **Intuition:**
  All these statements are **testable claims about population parameters**.
  - If it’s about mean → we test with **t-test or Z-test**.
  - If it’s about proportion → we test with **Z-test for proportions**.
  - If it’s about variance → we test with **Chi-square or F-test**.

✅ **Super Clear Recap:**

- **Hypothesis (general):** Any statement or guess about reality (daily life).
- **Statistical Hypothesis:** A statement about an **unknown population parameter** (mean, variance, proportion, distribution).
- **Why important?** Because statistics is all about verifying claims (company’s claim, product quality, machine efficiency, etc.) using sample evidence.

👉 Do you want me to now explain the **types of statistical hypothesis** (Null hypothesis H_0 vs Alternative hypothesis H_1) with strong intuition and examples (like your toothpaste claim)?



Definition of Hypothesis:

Any statement about any phenomenon is termed as hypothesis.

Definition of Statistical Hypothesis:

- Statistical hypothesis is a statement about population characteristic that can be tested on the basis of sample data.

- For example, a pious person will go to the heaven is not a statistical hypothesis since it can not be proved with the statistical data. It does not mean that it has no value. It is a faith.

In statistical tests of significance, two mutually exclusive hypotheses are to be used, these are null hypothesis and alternative hypothesis.

Great 👌 This page gives the **precise definitions** of hypothesis and statistical hypothesis, and introduces the two types: **Null (H_0)** and **Alternative (H_1)**. Let's unpack it step by step with intuition.

📖 Hypothesis & Statistical Hypothesis

---

1. Definition of Hypothesis (general)

> Any statement about any phenomenon is termed as a hypothesis.

- **Simple view:**
  A hypothesis = a *guess, claim, or assumption* about something.
  Example:
  - "Tomorrow will rain."
  - "This company is profitable."

👉 These are **general hypotheses**, not necessarily statistical.

---

2. Definition of Statistical Hypothesis

A statistical hypothesis is a statement about a **population characteristic** that can be tested using sample data.

- **Key difference:**
  - General hypothesis → Can be about *anything* (weather, religion, feelings, etc.).
  - Statistical hypothesis → Must be about a **population parameter** (mean μ, proportion p, variance σ^2, distribution).
- **Example in notes:**

  > *"A pious person will go to heaven"* → NOT a statistical hypothesis.

  - Why? Because it cannot be tested with **statistical data**.
  - It may be a matter of **faith**, but statistics only deals with measurable, testable claims.

✅ **Testable = Statistical Hypothesis**
❌ **Non-testable = Not Statistical**

---

3. Foundation for Hypothesis Testing

- In **statistical tests of significance**, we always frame **two competing hypotheses**:
  1. **Null Hypothesis (H_0)**
  2. **Alternative Hypothesis (H_1 or Ha)**
- **Mutually exclusive:** Only one can be true at a time.
  - If H_0 is true → we accept the claim.
  - If H_0 is rejected → we accept H_1.

---

4. Intuition with Examples

- Suppose a company claims toothpaste average weight = 140 gm.
  - **H_0 (Null):** $\mu = 140$ (claim is true).
  - **H_1 (Alternative):** $\mu \neq 140$ (claim is false).

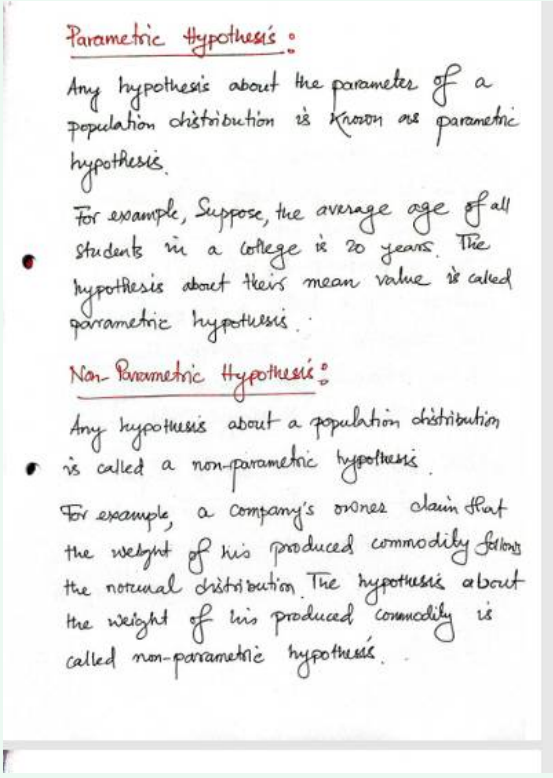
- Suppose a machine claims defect rate = 5%.
  - **H_0:** $p = 0.05$.
  - **H_1:** $p > 0.05$ (machine is worse).
    or H_1: $p < 0.05$ (machine is better).

👉 **H_0 = default belief (innocent until proven guilty).**
👉 **H_1 = suspicion / new claim.**

✅ **Super Clear Recap:**

- Hypothesis = any statement.
- Statistical hypothesis = statement about a population parameter, testable with sample data.
- Non-testable (like faith or metaphysics) = **not statistical**.
- In statistics, we always work with **two competing hypotheses**:
  - Null (H_0) = claim is true by default.
  - Alternative (H_1) = claim is false, something else is true.

👉 Do you want me to now move into **detailed explanation of Null Hypothesis vs Alternative Hypothesis (with one-tailed & two-tailed tests)** in your notes' style but with deeper intuition and real-life examples?


Parametric Hypothesis:

Any hypothesis about the parameter of a population distribution is known as parametric hypothesis.

For example, Suppose, the average age of all students in a college is 20 years. The hypothesis about their mean value is called parametric hypothesis.

Non-Parametric Hypothesis:

Any hypothesis about a population distribution is called a non-parametric hypothesis.

For example, a company's owner claim that the weight of his produced commodity follows the normal distribution. The hypothesis about the weight of his produced commodity is called non-parametric hypothesis.

Perfect 👍 now we are at **Parametric Hypothesis vs Non-Parametric Hypothesis**. Let's break it down in detail with **clear examples and intuition**.

📖 Parametric & Non-Parametric Hypotheses

---

1. Parametric Hypothesis

- **From notes:**

  > Any hypothesis about the **parameters** of a population distribution is known as a parametric hypothesis.

- **Meaning:**
  A parametric hypothesis directly involves **numerical population parameters** such as:
  - Mean (μ)
  - Proportion (p)
  - Variance (σ^2)

- **Example (from notes):**

  > "The average age of all students in a college is 20 years."

  - Here the **mean age $\mu = 20$** is the parameter.
  - This is **parametric hypothesis**.
- **Other Examples:**
  - $\mu = 50$ (mean marks in exam).
  - $p = 0.25$ (25% people prefer tea).
  - $\sigma^2 = 16$ (variance of weights).

👉 **Intuition:**
Think of "parametric" as **talking directly about numbers that describe the population**.

---

2. Non-Parametric Hypothesis

- **From notes:**

  > Any hypothesis about a **population distribution** is called a non-parametric hypothesis.

- **Meaning:**
  A non-parametric hypothesis is not about a specific numerical parameter, but about the **shape, pattern, or distribution** of the data.
- **Example (from notes):**

  > "The weight of produced commodities follows the normal distribution."

  - This is not about mean or variance, but about the **form of distribution** (Normal).
- **Other Examples:**
  - Data follows a **uniform distribution**.
  - Customer arrivals follow a **Poisson distribution**.
  - Income distribution is **not normal**.

👉 **Intuition:**
Think of "non-parametric" as **talking about the pattern/shape of data, not the exact numbers**.

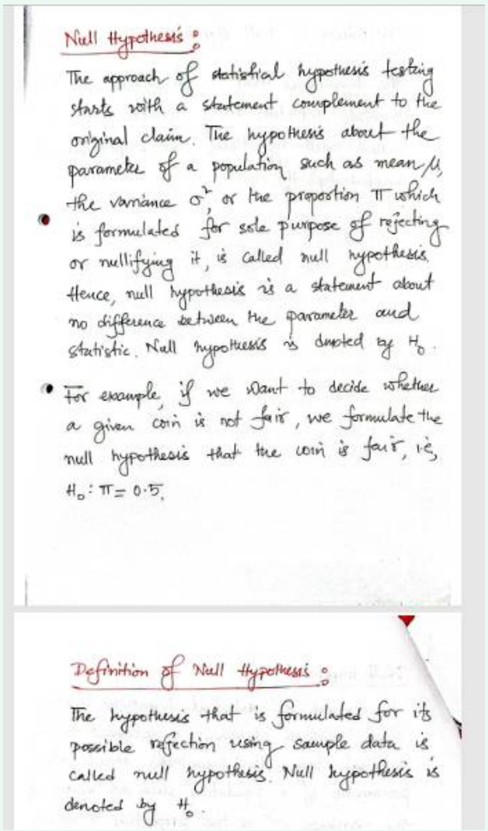
3. Quick Comparison Table

Aspect	Parametric Hypothesis	Non-Parametric Hypothesis
Focus	Numerical parameter (μ, p, σ^2)	Distribution form/shape
Example	"Average age = 20 years"	"Weights follow Normal distribution"
Common Tests Used	Z-test, t-test, F-test, χ^2 for variance	Chi-square goodness of fit, Kolmogorov-Smirnov test
Intuition	**About the numbers**	**About the pattern**

✅ **Super Clear Recap:**

- **Parametric Hypothesis** → Claim about population parameters (mean, variance, proportion).
- **Non-Parametric Hypothesis** → Claim about population distribution (Normal, Uniform, Poisson, etc.).

👉 Do you want me to now explain the **Null Hypothesis (H_0) vs Alternative Hypothesis (H_1) in detail, including one-tailed and two-tailed tests** with real-life intuition (like your toothpaste example)?



Null Hypothesis:

- The approach of statistical hypothesis testing starts with a statement complement to the original claim. The hypothesis about the parameter of a population such as mean μ, the variance σ^2, or the proportion π which is formulated for sole purpose of rejecting or nullifying it, is called null hypothesis. Hence, null hypothesis is a statement about no difference between the parameter and statistic. Null hypothesis is denoted by H_0.
- For example, if we want to decide whether a given coin is not fair, we formulate the null hypothesis that the coin is fair, i.e, $H_0: \pi = 0.5$.

Definition of Null Hypothesis:

The hypothesis that is formulated for its possible rejection using sample data is called null hypothesis. Null hypothesis is denoted by H_0.

Excellent 👍 you've reached the section on the **Null Hypothesis (H_0)**.
This is one of the most important ideas in hypothesis testing, so let's unpack it with **deep details + strong intuition + real-life examples**.

📖 Null Hypothesis (H_0)

---

1. From Notes (Restated)

- Hypothesis testing starts with a **statement complement to the original claim**.
- Hypothesis about a parameter (mean μ, variance σ^2, proportion p) formulated for the purpose of **rejecting or nullifying** is called the **Null Hypothesis**.
- Null hypothesis usually says:
  - *"There is no difference."*
  - *"The claim is true."*
- Denoted by **H_0**.

---

2. Intuition: “Innocent Until Proven Guilty”

- H_0 plays the role of **default assumption**.
- Just like in court:
  - A person is assumed **innocent** until proven guilty.
  - A claim (H_0) is assumed **true** until sample evidence shows strong reason to reject it.

👉 This makes hypothesis testing **fair and unbiased**.
You don’t reject a claim unless you have enough evidence.

---

3. Example (from your notes)

- **Problem:** Is a given coin fair?
- **Null hypothesis:**
  - H_0: $\pi = 0.5$ (probability of head = 0.5).
  - This means the coin is fair.
- **Alternative would be:**
  - H_1: $\pi \neq 0.5$ (coin is biased).

👉 Here, H_0 assumes fairness (the “no difference” case).
Testing = tossing the coin many times and checking if sample results strongly disagree with 0.5.

---

4. Formal Definition (from notes)

> The hypothesis that is formulated for its **possible rejection using sample data** is called Null Hypothesis.

- Denoted: $\mathbf{H_0}$

👉 Meaning: H_0 is not something we try to “prove true.”
It is something we **try to reject** if sample evidence is strong enough.

---

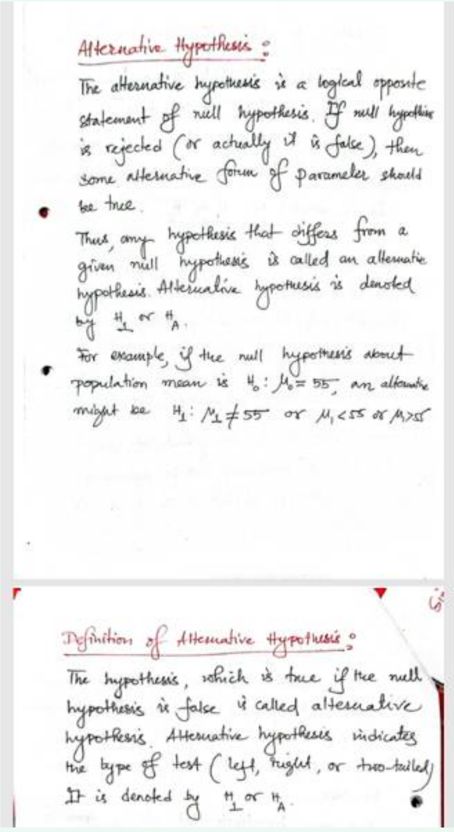
5. Everyday Examples of H_0

1. A medicine company claims:
   - H_0: "The drug has no effect."
   - H_1: "The drug is effective."
2. A battery company claims average life = 100 hours.
   - H_0: $\mu = 100$.
   - H_1: $\mu \neq 100$.
3. A student says they get 80% marks on average.
   - H_0: $\mu = 80$.
   - H_1: $\mu < 80$ (if we suspect they're exaggerating).

✅ Recap

- Null hypothesis (H_0) = the **default belief**.
- It usually represents **no difference / no effect / claim is true**.
- We don't prove H_0 true; we only check if sample evidence is strong enough to **reject it**.
- Example: "coin is fair" → H_0: $\pi = 0.5$.

👉 Next in your notes will be about the **Alternative Hypothesis (H_1)** and types (one-tailed vs two-tailed).
Would you like me to explain those **step by step with intuition + diagrams + real-world examples** (e.g., coin toss, toothpaste, medicine)?



Alternative Hypothesis :

The alternative hypothesis is a logical opposite statement of null hypothesis. If null hypothesis is rejected (or actually it is false), then some alternative form of parameter should be true.

- Thus, any hypothesis that differs from a given null hypothesis is called an alternative hypothesis. Alternative hypothesis is denoted by H_1 or H_A.
- For example, if the null hypothesis about population mean is $H_0 : \mu_0 = 55$, an alternative might be $H_1 : \mu_1 \neq 55$ or $\mu_1 < 55$ or $\mu_1 > 55$

Definition of Alternative Hypothesis :

The hypothesis, which is true if the null hypothesis is false is called alternative hypothesis. Alternative hypothesis indicates the type of test (left, right, or two-tailed) It is denoted by H_1 or H_A.

Excellent 🙌 You've now shared the **Alternative Hypothesis (H_1 or Ha)** section.
This is the "rival claim" to the **Null Hypothesis (H_0)**. Let's break it down in detail with **clear intuition, examples, and one-/two-tailed test connection**.

📖 Alternative Hypothesis (H_1 / Ha)

---

1. From Notes (Restated)

- Alternative hypothesis = **logical opposite of H_0**.
- If H_0 is rejected, then H_1 must be true.
- Any hypothesis that differs from H_0 is called H_1.
- Denoted: **H_1** or **Ha**.

---

2. Intuition

- Think of H_1 as the **suspicion or rival claim**.

- If H_0 is "the company's claim is true,"
  → H_1 is "No, the company's claim is false."

👉 Court analogy again:

- H_0: "Defendant is innocent."
- H_1: "Defendant is guilty."
- Jury can't say both. Rejecting one means accepting the other.

---

3. Example from Notes

- H_0: $\mu = 55$ (population mean is 55).
- Possible H_1:
  - H_1: $\mu \neq 55$ (not equal → two-tailed test).
  - H_1: $\mu < 55$ (less → left-tailed test).
  - H_1: $\mu > 55$ (greater → right-tailed test).

👉 So the **form of H_1 depends on the question**:

- Testing equality → two-tailed.
- Testing "less than" → left-tailed.
- Testing "greater than" → right-tailed.

---

4. Formal Definition (from your notes)

> The hypothesis which is true if the null hypothesis is false is called the Alternative Hypothesis.
> It indicates the type of test (left, right, or two-tailed).

---

5. Everyday Examples

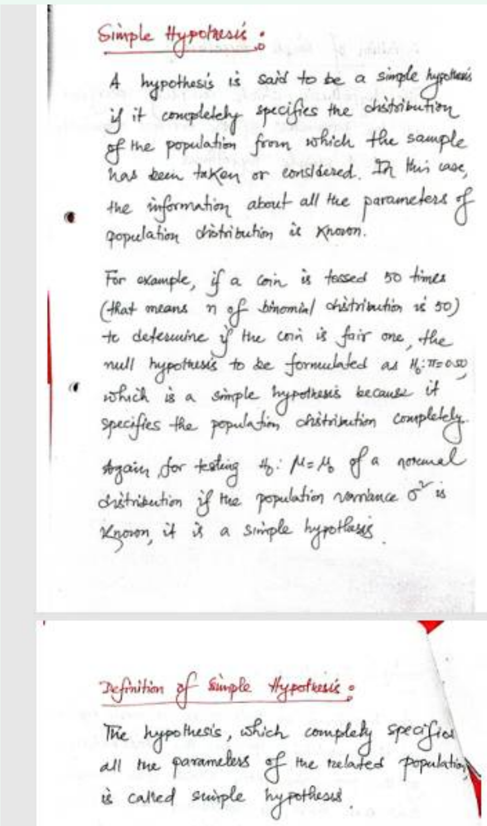
1. **Coin fairness test**
   - H_0: $p = 0.5$ (coin is fair).
   - H_1: $p \neq 0.5$ (coin is biased).

2. **Toothpaste weight**
   - H_0: μ = 140 g (claim true).
   - H_1: μ ≠ 140 g (claim false).
3. **Medicine effect**
   - H_0: μ = 0 (no effect).
   - H_1: μ > 0 (medicine works better than nothing).

✅ Recap

- **H_0 (Null):** Default claim, "innocent until proven guilty."
- **H_1 (Alternative):** Opposite claim, true if H_0 is false.
- **Types of H_1:**
  - μ ≠ value → **two-tailed test**.
  - μ < value → **left-tailed test**.
  - μ > value → **right-tailed test**.

👉 Do you want me to now explain the **One-tailed vs Two-tailed tests** with diagrams (bell curves, rejection regions) + intuition (real-life analogies like medicine effectiveness, coin toss, toothpaste claim)?



Simple Hypothesis:

A hypothesis is said to be a simple hypothesis if it completely specifies the distribution of the population from which the sample has been taken or considered. In this case, the information about all the parameters of population distribution is known.

For example, if a coin is tossed 50 times (that means n of binomial distribution is 50) to determine if the coin is fair one, the null hypothesis to be formulated as $H_0: \pi = 0.50$, which is a simple hypothesis because it specifies the population distribution completely.

Again for testing $H_0: \mu = \mu_0$ of a normal distribution if the population variance σ^2 is known, it is a simple hypothesis.

Definition of Simple Hypothesis:

The hypothesis, which completely specifies all the parameters of the related population is called simple hypothesis.

Great 👌 Now we're at **Simple Hypothesis**. This is an important classification because it tells us **how much detail the hypothesis gives about the population**. Let's break it down carefully with intuition and examples.

📖 Simple Hypothesis

---

1. From Notes (Restated)

- A hypothesis is called **simple** if it **completely specifies the distribution** of the population from which the sample is taken.
- In this case, **all parameters of the population distribution are known.**

---

2. Intuition

Think of a **simple hypothesis** as giving you a **clear, exact recipe** for the population.

- It doesn't leave anything vague.

- It tells you exactly which distribution and which parameters are assumed.

👉 In simple words:

- *"Everything is fixed and known except the sample data."*

---

3. Examples (from your notes + extra intuition)

Example 1: Coin Toss

- Suppose we toss a coin 50 times ($n = 50$ in Binomial distribution).
- Null hypothesis: $H_0: \pi = 0.5$ (probability of head = 0.5).
- This is **simple hypothesis** because:
  - Distribution (Binomial) is known.
  - Parameter ($\pi = 0.5$) is fully specified.

Example 2: Testing Normal Mean

- Suppose population is Normal(μ, σ^2).
- If we test $H_0: \mu = 40$, and σ^2 is already known.
- Then **distribution is completely specified** (Normal with $\mu = 40$, σ^2 known).
- This is also a **simple hypothesis**.

---

4. Formal Definition (from notes)

> A hypothesis which completely specifies all the parameters of the related population is called **Simple Hypothesis**.

---

5. Visual Intuition

Imagine you're cooking:

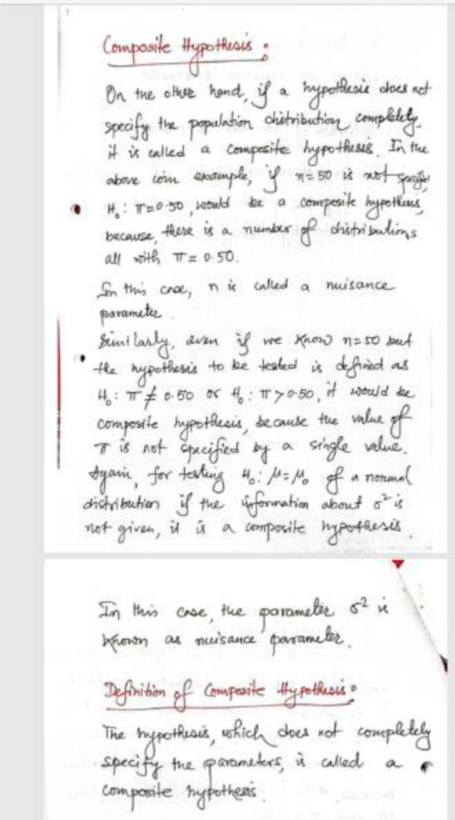
- If the recipe says: *"Use 2 cups of rice, 1 cup of lentils, and 3 cups of water"* → it's **simple** (everything is specified).

- That’s exactly what a simple hypothesis does → **fixes all parameters of the population**.

✅ **Super Clear Recap:**

- **Simple Hypothesis = completely specified distribution.**
- All parameters are known (e.g., π = 0.5, μ = 55 with known σ^2).
- Easy to test because nothing is uncertain except the sample.

👉 Do you want me to now continue with the **Composite Hypothesis** (the opposite case where parameters are not fully specified) — with intuition and comparison to Simple Hypothesis?



Composite Hypothesis :

On the other hand, if a hypothesis does not specify the population distribution completely, it is called a composite hypothesis. In the above coin example, if $n = 50$ is not specified, $H_0: \pi = 0.50$, would be a composite hypothesis, because there is a number of distributions all with $\pi = 0.50$.

In this case, n is called a nuisance parameter.

Similarly, even if we know $n = 50$ but the hypothesis to be tested is defined as $H_0: \pi \neq 0.50$ or $H_0: \pi > 0.50$, it would be composite hypothesis, because the value of π is not specified by a single value. Again, for testing $H_0: \mu = \mu_0$ of a normal distribution if the information about σ^2 is not given, it is a composite hypothesis.

In this case, the parameter σ^2 is known as nuisance parameter.

Definition of Composite Hypothesis :

The hypothesis, which does not completely specify the parameters, is called a composite hypothesis.

Perfect 👍 Now we’ve reached **Composite Hypothesis**, which is the opposite of **Simple Hypothesis**.
Let’s carefully explain this with **details, intuition, and examples**.

📖 Composite Hypothesis

1. From Notes (Restated)

- If a hypothesis does **not completely specify the population distribution**, it is called a **Composite Hypothesis**.
- In this case, some parameters are left **unknown** or not fixed.

2. Intuition

Think of it like a **recipe with missing instructions**:

- If a recipe says *"Use rice and lentils, but the amount can vary"* → you don't know the exact proportions.
- Similarly, in statistics, if parameters are not fully specified, the hypothesis is **composite**.

👉 Composite hypotheses describe a **range** or **set of possible distributions**, not just one fixed distribution.

3. Examples (from your notes + explained)

Example 1: Coin Toss

- H_0: $\pi = 0.5$, but **n (sample size)** is not specified.
- Then H_0 is **composite** because many different Binomial distributions are possible (Binomial with $n = 10$, $n = 20$, $n = 50$, ... all with $\pi = 0.5$).
- Here, **n is a nuisance parameter** (an extra unknown that complicates testing).

Example 2: Coin Bias

- Suppose $n = 50$ is fixed, but hypothesis is H_0: $\pi \neq 0.5$ or H_0: $\pi > 0.5$.
- This is also **composite** because π is not a single fixed value, it's a range (all values except 0.5, or all values > 0.5).

Example 3: Normal Distribution

- H_0: $\mu = 40$ for a Normal distribution, but variance σ^2 is not given.
- Then this is **composite** because distribution is not fully defined.
- Here, σ^2 is a **nuisance parameter**.

4. Formal Definition (from notes)

> A hypothesis which does not completely specify the parameters is called a **Composite Hypothesis**.

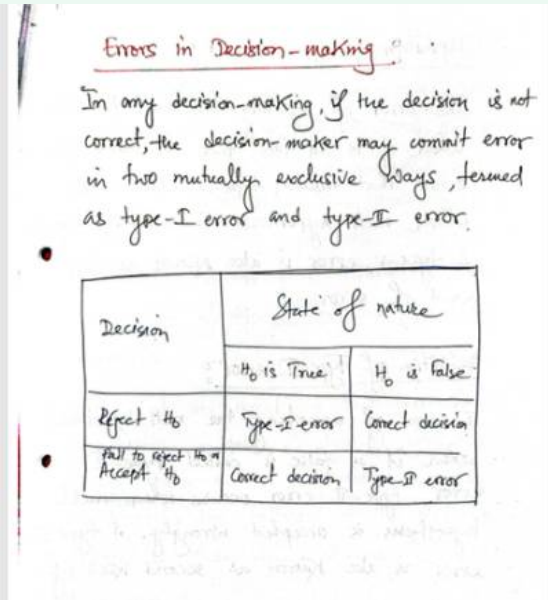
5. Simple vs Composite (Quick Comparison Table)

Aspect	Simple Hypothesis	Composite Hypothesis
Parameters	Fully specified	Not fully specified
Distribution	Only one possible distribution	Multiple possible distributions
Example	H_0: $\pi = 0.5$, $n = 50$	H_0: $\pi = 0.5$, but n not fixed OR σ^2 unknown
Testing	Easier (exact distribution known)	Harder (extra unknown → nuisance parameters)

✅ Super Clear Recap:

- **Simple Hypothesis** → fully specifies distribution (all parameters fixed).
- **Composite Hypothesis** → leaves parameters unknown or describes a range.
- Unknown parameters = **nuisance parameters**.
- Composite hypotheses are more common in real life, since often we don't know everything (like variance, exact sample size, etc.).

👉 Do you want me to now move on to explaining the **Steps of Hypothesis Testing (H_0, H_1, α, test statistic, decision rule, conclusion)** with a worked-out example (like your toothpaste or coin toss)?



Errors in Decision-making :

In any decision-making, if the decision is not correct, the decision-maker may commit error in two mutually exclusive ways, termed as type-I error and type-II error.

Decision	State of nature	
	H_0 is True	H_0 is False
Reject H_0	Type-I error	Correct decision
Fail to reject H_0 / Accept H_0	Correct decision	Type-II error

Perfect 👍 — this is a **very important part**: **Errors in Decision-Making** (Type I & Type II Errors).
Let's break it down clearly with **intuition, the table explained, and real-life examples**.

📖 Errors in Decision-Making (Hypothesis Testing)

---

1. From Notes (Restated)

- In hypothesis testing (decision-making), two kinds of errors are possible if our decision is wrong:
  - **Type I Error**
  - **Type II Error**

They are **mutually exclusive** → if one occurs, the other cannot happen at the same time.

---

2. Intuition (Courtroom Analogy again ⚖️)

- H_0 = "Accused is innocent."
- H_1 = "Accused is guilty."

Now two types of mistakes can happen:

- **Type I Error (False Alarm):**
  Convicting an innocent person (rejecting a true H_0).
- **Type II Error (Missed Detection):**
  Freeing a guilty person (accepting a false H_0).

👉 Hypothesis testing is just like court trials → we can make mistakes because we rely on **evidence (sample data)**, not 100% truth.

---

3. Table from Your Notes (Explained)

Decision	H_0 True	H_0 False
Reject H_0	❌ Type I Error (wrong)	✅ Correct Decision
Fail to reject H_0	✅ Correct Decision	❌ Type II Error (wrong)

---

4. Real-Life Examples

Example 1: Medical Test (Disease Detection)

- H_0: Patient is healthy.
- H_1: Patient is sick.
- **Type I Error:** Test says patient is sick (reject H_0), but actually healthy → **false positive**.
- **Type II Error:** Test says patient is healthy (fail to reject H_0), but actually sick → **false negative**.

👉 Type I = unnecessary worry/treatment.
👉 Type II = missed diagnosis, dangerous.

Example 2: Airport Security

- H_0: Passenger is not a terrorist.
- H_1: Passenger is a terrorist.
- **Type I Error:** Innocent passenger wrongly flagged.

- **Type II Error:** Terrorist goes undetected.

Example 3: Coin Toss

- H_0: Coin is fair (p = 0.5).
- H_1: Coin is biased.
- **Type I Error:** Rejecting fairness when coin is actually fair.
- **Type II Error:** Accepting fairness when coin is actually biased.

---

5. Controlling Errors

- **Type I Error (α):**
  - Probability of making a Type I error is controlled by **significance level α** (usually 0.05 or 5%).
  - Smaller α → less chance of false alarm.
- **Type II Error (β):**
  - Probability of Type II error depends on sample size and test power.
  - Larger sample → less chance of missing the truth.

✅ Super Clear Recap:

- **Type I Error (α):** Rejecting a true H_0 → false alarm.
- **Type II Error (β):** Failing to reject a false H_0 → missed detection.
- Controlled by significance level (α) and test design (power = 1 – β).

👉 Do you want me to now explain **Level of Significance (α), Power of Test, and their relation to Type I & Type II errors** step by step with intuition (like traffic police checking licenses, medical testing, etc.)?

Definition of Type-I error:

The error of rejecting the null hypothesis when it is in fact true is called type-I error. That means, type-I error occurs when null hypothesis is wrongly rejected. A type-I error is also known as first kind of error.

Definition of Type-II error:

The error of accepting the null hypothesis when it is false is called type-II error. Type-II error occurs when null hypothesis is accepted wrongly. A type-II error is also known as second kind of error.

A type-I error is often considered to be more serious, and therefore more important to avoid than a type-II error. The hypothesis test procedure is therefore adjusted so that there is a guaranteed 'low' probability of rejecting the null hypothesis wrongly. This probability is never 0. While the exact probability of type-II error is generally unknown.

If we do not reject the null hypothesis, it may still be false (a type-II error) as the sample may not be representative enough to identify the falseness of the null hypothesis (especially if the truth is very close to hypothesis)

For any given set of data, type-I & type-II errors are inversely related; the smaller the

Excellent 👍 you've now reached the **detailed explanation of Type I and Type II Errors**. Let's unpack it in detail with **definitions, seriousness, and intuition**.

📖 Type I & Type II Errors

---

1. Definition of Type I Error (α)

- **From notes:**

  > The error of rejecting H_0 when it is actually true.

- **Also called:** First kind of error.

👉 **Intuition:**
This is like a **false alarm** → saying something is wrong when in fact it's fine.

- Example: Declaring a fair coin biased.
- Example: Diagnosing a healthy patient as sick (false positive).

---

2. Definition of Type II Error (β)

- **From notes:**

  > The error of accepting H_0 when it is actually false.

- **Also called:** Second kind of error.

👉 **Intuition:**
This is like a **missed detection** → failing to notice a problem when it exists.

- Example: Declaring a biased coin as fair.
- Example: Saying a sick patient is healthy (false negative).

---

3. Which Error is More Serious?

- **Notes say:**
  - Type I Error is often considered **more serious**.
  - Therefore, hypothesis testing procedures are designed to keep the probability of Type I Error (α) **very low** (commonly 5% or 1%).

👉 **Why?**

- Convicting an innocent person (Type I) is considered worse than letting a guilty go free (Type II).
- Giving unnecessary treatment to a healthy person is worse than missing mild cases sometimes.

---

4. Probability of Errors

- **Type I Error (α):**
  - Probability is fixed and controlled at the beginning of test (e.g., $\alpha = 0.05$).
  - This is called the **Level of Significance**.
- **Type II Error (β):**
  - Probability depends on sample size and effect size.
  - Exact value is often unknown.
  - Power of test = $1 - \beta$ (probability of correctly rejecting a false H_0).

---

5. Relationship Between α and β

- From your notes:

  > For any given set of data, Type I and Type II errors are inversely related.

  - Lowering α (stricter test) → increases β (harder to detect false H_0).
  - Increasing α (looser test) → decreases β.

👉 **Balance needed:**
We can't minimize both at once. So we **fix α small** and then try to reduce β by increasing sample size.

---

6. Recap with Intuition

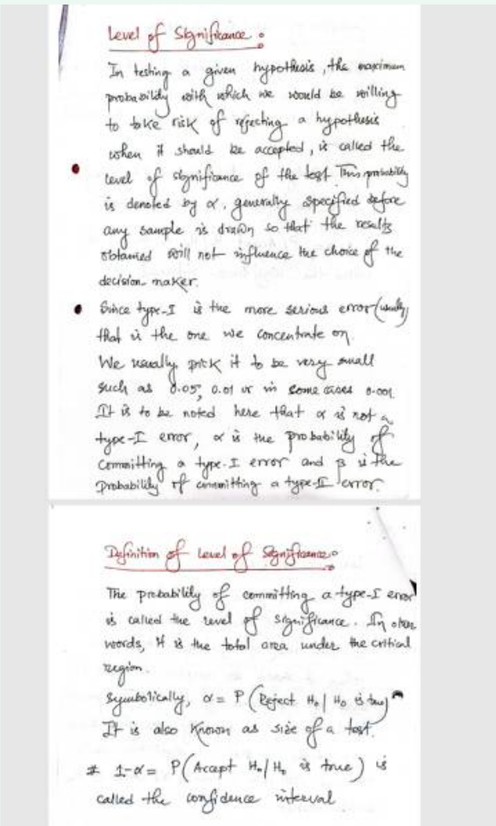
- **Type I Error (α):** Rejecting a true H_0 → *false alarm.*
- **Type II Error (β):** Accepting a false H_0 → *missed detection.*
- **Control:** α is fixed first (significance level), then reduce β by larger samples or better test design.
- **Trade-off:** Small α → larger β, unless sample size increases.

✅ **Everyday Analogy:**
Imagine a fire alarm:

- **Type I Error (false alarm):** Alarm rings when there is no fire.

- **Type II Error (missed detection):** No alarm when there is a real fire.
- We usually design alarms to **minimize false alarms (α)** but also try to make them sensitive enough to detect actual fires (reduce β).

👉 Do you want me to now go into **Level of Significance (α), Power of Test (1 – β), and how they are practically chosen** with intuitive examples (like medical testing and court trials)?



Level of Significance :

In testing a given hypothesis, the maximum probability with which we would be willing to take risk of rejecting a hypothesis when it should be accepted, is called the

• level of significance of the test. This probability is denoted by α, generally specified before any sample is drawn so that the results obtained will not influence the choice of the decision maker.

• Since type-I is the more serious error (usually) that is the one we concentrate on. We usually pick it to be very small such as 0.05, 0.01 or in some cases 0.001. It is to be noted here that α is not a type-I error, α is the probability of committing a type-I error and β is the probability of committing a type-II error.

Definition of Level of Significance :

The probability of committing a type-I error is called the level of significance. In other words, it is the total area under the critical region.

Symbolically, $\alpha = P(\text{Reject } H_0 \mid H_0 \text{ is true})$

It is also known as size of a test.

$1-\alpha = P(\text{Accept } H_0 / H_0 \text{ is true})$ is called the confidence interval

Perfect 👍 You've now shared the section on **Level of Significance (α)**.
This connects directly to **Type I Error**. Let's carefully explain with **intuition, examples, and formulas**.

📖 Level of Significance (α)

---

1. From Notes (Restated)

- Level of Significance = the **maximum probability of risk** we are willing to take of rejecting H_0 when it is actually true.
- Denoted by **α**.
- Chosen **before collecting data** (so that results don't bias the decision).
- Typical values: **0.05, 0.01, 0.001**.

---

2. Intuition

Think of α as the **strictness of the judge** (or the threshold for evidence).

- If α = 0.05 → we allow **5% chance of wrongly rejecting H_0**.
- If α = 0.01 → we allow only **1% chance of wrongly rejecting H_0**.
- Smaller α → harder to reject H_0, meaning we require **stronger evidence**.

👉 So α = "How much doubt are we willing to tolerate before rejecting the default claim?"

---

3. Why focus on α (Type I Error)?

- Notes say: Type I Error is more serious, so we control it carefully.
- Example: In medicine testing → giving medicine to a healthy person (false alarm) may have dangerous side effects, so α must be low.

---

4. Mathematical Definition

- **Level of Significance (α):**

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ is true})$$

  → Probability of committing a Type I Error.

- **Confidence Level:**

$$1 - \alpha = P(\text{Accept } H_0 \mid H_0 \text{ is true})$$

  → Probability of making the correct decision when H_0 is true.
  This is also called the **Confidence Interval**.

5. Example: Coin Toss

- H_0: Coin is fair ($p = 0.5$).
- $\alpha = 0.05$ → we allow only 5% chance of wrongly rejecting fairness.
- So rejection region = outcomes so extreme they happen only 5% of the time under H_0 (e.g., too many heads or too many tails).

---

6. Example: Medicine Trial

- H_0: Medicine has no effect.
- $\alpha = 0.01$ → only if probability of getting the observed effect under H_0 is less than 1%, we reject H_0.
- Meaning: we need **very strong evidence** before approving the medicine.

✅ Recap

- **Level of Significance (α):** Maximum risk of rejecting H_0 wrongly (Type I Error).
- Common α values: 0.05 (5%), 0.01 (1%), 0.001 (0.1%).
- Smaller α → stronger evidence required.
- Confidence Level = $1 - \alpha$.

👉 Do you want me to now explain the **Power of a Test (1 – β)** and how it relates to α, with examples (medical test, security checks, coin toss)?

OpenAI

Definition of Power of a Test :

The complement of the probability of type-II error is called the power of a test. That means, the probability of rejecting a false null hypothesis is the power of a test.

In other words, the probability of correct decision is the power of a test. Symbolically

Power of a test $1-\beta = 1 - P(\text{Accept } H_0 / H_0 \text{ is false})$

$= P(\text{Reject } H_0 / H_0 \text{ is false})$

Interpretation of level of Significance :

Generally, a significance level of 0.05 or 0.10 is considered, although other values are also used. Thus, if the level of significance is 0.05, it will mean that there are about 5 samples out of 100 that would direct to reject the hypothesis when it should be actually accepted.

So $(1-0.05) = 0.95$ is the probability of accepting null hypothesis when it is true, i.e., there is 95% confidence in taking the right decision.

In such case it is said that the hypothesis has been rejected at 5% level of significance, which again means that the probability of wrong decision is 0.05.

Excellent 👍 you've now reached **Power of a Test** and **Interpretation of Level of Significance (α)**.
These two are directly linked with **Type I & Type II Errors**. Let's carefully explain with **step-by-step intuition**.

📖 Power of a Test

---

1. From Notes (Restated)

- **Power of a test = 1 – β**
- β = probability of Type II Error (failing to reject H_0 when it is false).
- So, **Power = probability of rejecting H_0 when H_0 is false**.
- In other words → **probability of making the correct decision when the null hypothesis is false**.

---

2. Intuition

Think of a medical test:

- H_0: Patient is healthy.
- H_1: Patient is sick.
- **β (Type II Error):** Test fails to detect the disease → false negative.
- **Power (1 – β):** Test correctly detects the disease when patient is sick → true positive.

👉 Power is like the **strength of the test** → how good it is at catching false claims.

---

3. Formula

$$\text{Power of a Test} = 1 - \beta = P(\text{Reject } H_0 \mid H_0 \text{ is false})$$

---

4. Example

- If β = 0.20 → 20% chance of missing the truth.
- Then Power = 1 – 0.20 = 0.80 → 80% chance of correctly detecting the truth.
- Good tests usually aim for **power ≥ 0.80**.

📖 Interpretation of Level of Significance (α)

---

1. From Notes (Restated)

- Common α values: **0.05 or 0.10**.
- If α = 0.05 → means about **5 times in 100 trials**, we will wrongly reject a true H_0.
- So, **95% of the time we are confident** we make the right decision when H_0 is true.
- This is called the **confidence level (1 – α)**.

---

2. Intuition

If α = 0.05:

- We accept a **5% risk of false alarm** (Type I Error).
- Confidence level = 95%.

👉 So when we say *"We reject H_0 at 5% significance level"*, we mean:

- We're **95% confident** in our decision.
- There's still a **5% chance we could be wrong**.

---

3. Example

- Suppose we test toothpaste weight claim μ = 140 gm at α = 0.05.
- If p-value < 0.05 → reject H_0.
- Meaning: we are 95% confident the toothpaste weight is not 140 gm.
- But still, there's a 5% risk that we made a mistake (rejected a true claim).

✅ Super Clear Recap

- **Power of a Test (1 – β):** Ability to correctly reject false H_0 (test strength). Should be ≥ 80%.
- **Level of Significance (α):** Maximum risk of rejecting a true H_0 (false alarm). Usually 5%.
- **Confidence Level (1 – α):** Confidence in our decision (e.g., 95%).

👉 Do you want me to now move on to the **steps of hypothesis testing procedure** (H_0, H_1, α, test statistic, decision, conclusion) and work out a **full example** (like the toothpaste claim or coin toss)?

One-tailed & two-tailed test:

A one-tailed test is a test, which is concerned about possible deviation of the value of the parameter in only one direction from the specified value defined in the null hypothesis, while a two-tailed test is a test, which is concerned about the possible deviation of the parametric value in both directions.

These are also called a one-sided alternative or two-sided alternatie.

In this case, the parameter can take any value other than the value specified by null hypothesis.

For example, $H_0: \mu = 0$ against $H_1: \mu \neq 0$ is a two-tailed test, while $H_0: \mu = 0$, against $H_1: \mu > 0$ or $H_0: \mu = 0$ against $H_1: \mu < 0$ is a one-tailed test.

Excellent 👍 this is the section on **One-tailed & Two-tailed Tests**.
This is very important because it decides **where we look for evidence against H_0**.
Let's go through it step by step with **intuition, diagrams (conceptual), and examples**.

📖 One-tailed & Two-tailed Tests

1. From Notes (Restated)

- **One-tailed test:** Concerned about deviation of the parameter in **only one direction** (greater or less).
- **Two-tailed test:** Concerned about deviation of the parameter in **both directions** (greater or less).
- Also called **one-sided alternative** or **two-sided alternative**.

---

2. Intuition (Simple Picture in Mind)

Think of H_0 as "standing in the middle."

- **Two-tailed test:**
  You check **both sides** → could be too high or too low.
  - Example: "Is the coin fair?" ($p = 0.5$).
  - If too many heads **or** too many tails → reject H_0.
- **One-tailed test:**
  You only check **one side** → only higher or only lower matters.
  - Example: "Is this medicine better than the old one?"
  - We only care if new mean > old mean, not if it's less.

👉 Easy analogy:

- **Two-tailed = checking both exits of a house.**
- **One-tailed = guarding only one exit.**

---

3. Examples (from notes + intuition)

- **Two-tailed test**
  - H_0: $\mu = 50$
  - H_1: $\mu \neq 50$
  - Reject if μ is either much larger or much smaller than 50.
- **One-tailed test (Right-tailed)**

  - H_0: μ = 50
  - H_1: μ > 50
  - Reject only if μ is significantly greater.
- **One-tailed test (Left-tailed)**
  - H_0: μ = 50
  - H_1: μ < 50
  - Reject only if μ is significantly smaller.

---

4. Visualization (Imagine a Bell Curve 📉)

- **Two-tailed test (α split in two):**
  - Rejection regions are at both ends of the curve.
  - Each tail has α/2.
  - Example: α = 0.05 → 0.025 in each tail.
- **One-tailed test:**
  - Rejection region is only in **one tail**.
  - Example: α = 0.05 → entire 0.05 in one tail.

---

5. Everyday Life Examples

- **Two-tailed:**
  Testing if a scale is accurate.
  - H_0: Scale shows correct weight.
  - H_1: Scale shows wrong weight (too heavy or too light).
- **One-tailed:**
  Testing if a new fertilizer increases crop yield.
  - H_0: Yield = old yield.
  - H_1: Yield > old yield.
    (We only care if it's better, not worse.)

✅ Super Clear Recap:

- **One-tailed test:** Evidence in one direction only (μ > or μ <).
- **Two-tailed test:** Evidence in both directions (μ ≠).
- Choice depends on research question:
  - If you care about any difference → two-tailed.
  - If you care about only increase or only decrease → one-tailed.

👉 Do you want me to now show you **step-by-step procedure of hypothesis testing (H_0, H_1, α, test statistic, decision, conclusion)** with a **worked-out numerical example** (like coin toss or toothpaste claim)?

A left-tailed test:

When the rejection region is in the left tail of the distribution of the test statistic, the test is called a left-tailed test. If the null hypothesis is $H_0: \mu_0 = 0$ then the alternative hypothesis will be $H_1: \mu < 0$.

A right-tailed test:

When the rejection region is in the right tail of the distribution of the test statistic, the test is called a right-tailed test. If the null hypothesis is $H_0: \mu_0 = 0$ then the alternative hypothesis will be $H_1: \mu > 0$.

Alternative hypotheses defined in the above two tests are called one-sided alternatives.

A two-tailed test:

When the rejection region is equally divided in the left and right tails of the distribution of the test statistic, the test is called a two-tailed test. The alternative hypothesis defined in this test is called two-sided alternatives. If the null hypothesis is $H_0: \mu_0 = 0$, then the two sided alternative hypothesis is defined by $H_1: \mu_0 \neq 0$.

In the test of significance, the decision whether the use of a one-tailed or a two-tailed test is appropriate depends on the objective of the test, i.e., It is chosen on the basis of the direction of claimed deviation of the parameter.

Suppose that a manufacturer of ballpoint pen, whose machine produces on average 1000 pens per hour, is planning to purchase a new machine

The authority will not buy a new machine unless it is definitely proved as superior. For this he would test the hypothesis that the new machine is no better than the existing machine or similar to the machine now available. in the market against the alternative that the new machine is superior with respect to the hourly average production of the existing machine.

In other words, it is required to test the null hypothesis $H_0: \mu = 1000$ against the alternative $H_1: \mu > 1000$ and buy the new machine only if the null hypothesis is rejected. Such an alternative test will result in a one-tailed test with the critical region in the right tail.

Excellent 👍 You've now shared the detailed explanation of **Left-tailed, Right-tailed, and Two-tailed Tests**, along with a **real-life example (ballpoint pen machine)**. Let's break this down step by step with **intuition + examples + reasoning**.

📖 One-tailed and Two-tailed Tests

---

1. Left-tailed Test

- **From notes:**
  - If rejection region is in the **left tail** of the distribution → left-tailed test.
  - Example: H_0: $\mu = 0$ vs H_1: $\mu < 0$.
- **Intuition:**
  We are testing if the parameter is **smaller than expected**.

- **Real-life example:**
  - Company claims a rope can hold **50 kg**.
  - H_0: $\mu = 50$.
  - H_1: $\mu < 50$.
  - If average strength is too low, reject H_0 (safety issue).

---

2. Right-tailed Test

- **From notes:**
  - If rejection region is in the **right tail** of the distribution → right-tailed test.
  - Example: H_0: $\mu = 0$ vs H_1: $\mu > 0$.
- **Intuition:**
  We are testing if the parameter is **greater than expected**.
- **Real-life example:**
  - Testing if new medicine works better.
  - H_0: $\mu = 0$ (no improvement).
  - H_1: $\mu > 0$ (positive improvement).
  - Only large positive evidence → reject H_0.

---

3. Two-tailed Test

- **From notes:**
  - If rejection region is **split equally** between both tails → two-tailed test.
  - Example: H_0: $\mu = 0$ vs H_1: $\mu \neq 0$.
- **Intuition:**
  We are testing if the parameter is **different in either direction** (could be higher or lower).
- **Real-life example:**
  - Testing if a scale is accurate.
  - H_0: μ = correct weight.

- H_1: $\mu \neq$ correct weight (could be over- or under-weighing).

---

4. Which Test to Use?

- Depends on the **research question**:
  - If we only care about **improvement (increase)** → right-tailed.
  - If we only care about **decrease (shortfall)** → left-tailed.
  - If we care about **any difference** → two-tailed.

---

5. Real-life Example from Notes: Ballpoint Pen Machine

- Old machine produces on average **1000 pens per hour**.
- Company is considering a **new machine**.
- **Null Hypothesis (H_0):** $\mu = 1000$ (new machine is no better).
- **Alternative Hypothesis (H_1):** $\mu > 1000$ (new machine is better).
- Decision rule:
  - Only if evidence shows new machine's average output is **significantly higher**, we buy it.
  - If it's equal or lower → don't buy.

👉 This is a **Right-tailed Test** because we only care about the **increase**.

✅ **Super Clear Recap:**

- **Left-tailed test:** Checks if $\mu <$ value. (Reject if left extreme).
- **Right-tailed test:** Checks if $\mu >$ value. (Reject if right extreme).
- **Two-tailed test:** Checks if $\mu \neq$ value. (Reject if either extreme).
- Choice depends on **what direction matters in decision-making**.

👉 Do you want me to now work out a **step-by-step numerical example** of a **one-tailed and two-tailed test** (with Z-test or t-test) so you see the full calculation and conclusion process?